- **Modules**:

Modules are files containing Python code that may include functions, classes or variables. Importing modules allows you to access their functionality in your own scripts.

Modules allow you to divide your program in to individual files that can be reused as needed.

1. Standard Modules: (*import math*)
2. Creating and importing your own modules

- **Packages**:

A package is a folder that contains one or more Python modules. It must also contain a special module called *__init__.py*

The *__init__.py* module does not need to contain any code. It only needs to exist so that Python recognizes the package folder as a Python package.

```
mypackage/
│
├── __init__.py
├── module1.py
└── module2.py
```

Modules from the package are imported using the syntax:

*import <package_name>.<module_name> as <alias>*

*Or*

*from <package_name> import <module_name> as <alias>*

- **Sub-packages:**

A package nested inside another package is called sub-package.

```
mypackage/
│
├── mysubpackage/
│   ├──__init__.py
│   └── module3.py
│
├── __init__.py
├── module1.py
└── module2.py
```

Modules from sub-packages are imported using syntax.

*import <package_name>.<subpackage_name>.<module_name> as <alias>*

*Or*

*from <package_name>.<subpackage_name> import <module_name> as <alias>*

## Note:

Subpackages are great practice for organizing your code inside the very large packages. They help keep the folder structure of a package clean and organized.

However deeply nested subpackages introduce long dotted module names. You can imagine how much typing it would take to import the module from a subpackage of a subpackage of a subpackage of a package.

It is a good practice to try and keep your subpackages at most one or two levels deep.

- ***Module and Packages (Task):***

1. Create a project folder called *package_exercise/ ,* now create a package called *helper/* with three modules: *__init__.py, string.py and maths.py.*

    - In the *string.py* module, add a function called *shout()* that takes a single string parameter and returns a new string with all of the letters in uppercase.

    - In the *maths.py* module, add a function called *area()* that take two parameters called *length* and *width* and return their product *length x width*.

2. In the root project folder, create a module called *main.py* that import *shout()* and *area()* functions. Use the *shout() and area()* functions to print the following output:

    *THE AREA OF A 5-BY-8 RECTANGLE IS 40*